# When a Number Rule Stops Working
## Answer Key

Grades 4–5 Mathematics

### Quick Answers

1. (a) correct 4th term = 22, (b) correct 5th term = 28, —
2. 27
3. (a) first term that is not prime = 15, —
4. (a) the counterexample = 6, —
5. (a) correct 5th term = 25, —
6. (a) 10th term of pattern A = 32, (b) 10th term of pattern B = 56, (c) comparison sign = $<$, —
7. (a) 6th term = 243, —
8. (a) the counterexample = 12, —

### What is verified

1 of 8 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 1, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

### Curriculum

**Level:** Grades 4–5 Mathematics
4.OA.B.4 – *problems 2, 4, 8*
4.OA.C.5 – *problems 1, 3, 5, 6, 7*
*Difficulty 1–5 of 5 across 18 tagged checks.*

### Common wrong answers

1. *If they got 24: added 8 instead of 6 to reach the fourth term*
1. *If they got 30: carried the wrong fourth term forward and added 6 to it*
3. *If they got 19: skipped past the fourth term and named the fifth term, which is prime*
5. *If they got 13: assumed the step stays 3 forever instead of growing*

**1.** Ravi is writing a pattern that starts at 4 and adds 6 each time. He wrote 4, 10, 16, 24, 30. (a) Correct 4th term. (b) Correct 5th term. (c) Explain what Ravi did wrong.

**Solution:** The rule is “start at 4, add 6”. Term by term: 4, $4 + 6 = 10$, $10 + 6 = 16$, $16 + 6 = 22$, $22 + 6 = 28$.

**(a)** The third term 16 is correct; the fourth should be $16 + 6$. $\boxed{22}$

**(b)** Adding 6 once more gives $\boxed{28}$

**(c) — instructor-judged.** Look for a response that says the step from the third term to the fourth was 8 instead of 6, and that everything after it is therefore 2 too large. Naming the wrong step size is what earns full credit; simply rewriting the corrected list is not an explanation.

*Common wrong answers:* 24 and 30 — the terms produced by that single 8-step and then carried forward.

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**2.** Mia says: “Every whole number that ends in the digit 7 is prime.” Find the smallest number greater than 7 that ends in 7 and is not prime, and write it as a product of two whole numbers, neither of them 1.

**Solution:** Test the numbers ending in 7 in order. 17 is prime. 27 is not: $27 = 3 \times 9$, so it has factors other than 1 and itself.

The smallest counterexample is therefore 27, written as a product. $\boxed{3 \times 9 = 27}$

Mia’s rule was built from a handful of small cases that happened to be prime. Ending in 7 says nothing about a number’s factors.

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**3.** Ben writes a pattern that starts at 3 and adds 4 each time, checks the first three terms, sees each is prime, and says “Every number in my pattern is prime.” (a) Write the first term that is not prime. (b) Explain why checking three terms was not enough.

**Solution:** The pattern runs $3, 7, 11, 15, 19, \ldots$ The first three terms really are prime. The fourth term is $3 + 3 \times 4$.

**(a)** $15 = 3 \times 5$, so it is not prime. $\boxed{15}$

**(b) — instructor-judged.** Look for a response that says a claim about *every* term has to hold for every term, so three confirming cases prove nothing — and that one counterexample is enough to defeat the claim.

*Common wrong answer:* 19 — reached by skipping past the fourth term; 19 is prime, so it is not a counterexample at all.

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**4.** Omar says: “Every multiple of 3 is odd. Look at 3, 9, 15 and 21.” (a) Write the smallest multiple of 3 that is greater than 3 and is even. (b) Explain why Omar’s examples led him to a rule that is false.

**Solution (a):** Multiples of 3 in order: 3, 6, 9, 12. The second one is $3 \times 2$, which is even. $\boxed{6}$

**(b) — instructor-judged.** Look for a response that notices every example Omar picked was 3 multiplied by an *odd* number, so the products came out odd. Multiplying 3 by an even number gives an even multiple, and his sample never contained one.

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**5.** A pattern begins 1, 4, 9, 16, ... Toby sees the first step is +3 and decides the rule is “add 3 every time”. He writes 13 as the 5th term. (a) Write the correct 5th term. (b) Explain why Toby’s rule does not fit this pattern.

**Solution:** The terms are the square numbers: $1 \times 1$, $2 \times 2$, $3 \times 3$, $4 \times 4$. The steps between them are +3, +5, +7 — growing, not constant.

**(a)** The 5th term is $5 \times 5$. $\boxed{25}$

**(b) — instructor-judged.** Look for a response that says the step size itself changes (3, then 5, then 7, then 9), so an add-3 rule matches only the very first step.

*Common wrong answer:* 13 — what "add 3 every time" actually produces.

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**6.** Priya says: "A pattern that starts with a bigger number always has bigger terms later on." Pattern A starts at 5 and adds 3; pattern B starts at 2 and adds 6. (a) 10th term of A. (b) 10th term of B. (c) Write ¡ or ¿ between them. (d) Explain why Priya's rule is wrong.

**Solution:** To reach the 10th term, take 9 steps from the start.

**(a)** Pattern A: $5 + 9 \times 3 = 5 + 27$. $\boxed{32}$

**(b)** Pattern B: $2 + 9 \times 6 = 2 + 54$. $\boxed{56}$

**(c)** Comparing the two: $\boxed{32 < 56}$

**(d)** — **instructor-judged.** Look for a response that identifies the step size as the deciding factor: pattern B starts three lower but gains twice as much on every step, so it overtakes pattern A and stays ahead. A response that only says "Priya is wrong" is incomplete.

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**7.** A pattern starts at 1 and each term is 3 times the one before. Dana says: "Every term is odd, so the rule must be add an even number each time." (a) Write the 6th term. (b) Explain why Dana's rule cannot be the rule for this pattern.

**Solution:** The terms are $1, 3, 9, 27, 81, \ldots$ — each one 3 times the last, so the 6th term is 3 multiplied by itself 5 times.

**(a)** $3 \times 3 \times 3 \times 3 \times 3 = 243$. $\boxed{243}$

**(b)** — **instructor-judged.** Look for a response that separates a shared *property* from the *rule*: the terms are all odd, but that is a consequence of multiplying odd numbers, not the rule itself. An "add an even number" rule with a fixed step cannot produce steps of 2, 6, 18, 54.

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**8.** Ines says: "A number divisible by 2 and by 3 is always divisible by 6. So a number divisible by 2 and by 4 must always be divisible by 8." (a) Write the smallest whole number greater than 8 that is divisible by both 2 and 4 but not by 8. (b) Explain why the first rule works but the second one does not.

**Solution (a):** Check the multiples of 4 above 8: $12 = 4 \times 3$. It is divisible by 2 and by 4, and $12 \div 8$ leaves a remainder, so it is not divisible by 8. $\boxed{12}$

**(b)** — **instructor-judged.** Look for a response built on shared factors: 2 and 3 have no factor in common, so a number carrying both must carry their product. But 4 already contains the 2, so "divisible by 2" adds nothing new once "divisible by 4" is known, and nothing forces a second factor of 2 beyond the one 4 supplies.